FIRST ABU DHABI BANK

Group Risk Management

MODEL RISK MANAGEMENT FRAMEWORK

AI / ML and Quantitative Model Governance Standards

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1. Framework Purpose and Scope

This Model Risk Management (MRM) Framework establishes FAB's enterprise-wide standards for the identification, measurement, management, and governance of model risk. It applies to all quantitative models, AI/ML systems, and automated decision-making tools used within FAB Group that are classified as material under Section 2 of this Framework.

The Framework is aligned with internationally recognised standards, including the US Federal Reserve SR 11-7 guidance on model risk management and the CBUAE Circular No. 2024/BSE/047 on AI/ML governance.

> **SCOPE:** This Framework explicitly covers AI-based pricing recommendation systems, including large language model (LLM)-based agents used to propose credit facility pricing, as these systems generate quantitative outputs used in material business decisions.

2. Model Risk Classification

FAB classifies all models into three tiers based on materiality and the potential consequences of model error:

Tier	Criteria	Examples	Validation Requirement
Tier 1 (High)	Direct P&L impact or credit decision influence above AED 10M; regulatory capital models; pricing AI agents	IFRS 9 impairment models · Credit pricing AI · Stress testing models	Full independent validation before deployment; annual
Tier 2 (Medium)	Indirect P&L impact; operational models with significant throughput	Customer segmentation · behavioural scoring models	Targeted validation; biennial
Tier 3 (Low)	Minimal impact; administrative or back-office tools	Document classification · internal scheduling tools	Self-certification by owner

3. Model Evidence Pack Requirements

For every Tier 1 and Tier 2 model deployed in production, a complete Model Evidence Pack (MEP) must be maintained and updated following any material model change. The MEP is the primary audit document for model risk purposes and must be made available to internal audit, CBUAE examiners, and external auditors upon request.

3.1 Mandatory MEP Components — All Models

1. Model Overview: Name, version, unique model ID, purpose, business function, date of initial development, and current deployment status.
2. Model Design Documentation: Theoretical basis, mathematical formulation, assumptions and limitations, data sources, and feature engineering approach.
3. Development and Training Data: Data dictionary, data sources, data quality assessment, training period, and any data preprocessing or transformations applied.

4. Performance Metrics: In-sample and out-of-sample performance; benchmark comparison against challenger models or expert judgment; results of cross-validation.
5. Validation Record: Independent validation findings, identified weaknesses, management responses, and outstanding remediation actions.
6. Model Limitations and Known Failure Modes: Documented scenarios in which the model may perform poorly or produce unreliable outputs, and the controls in place to mitigate such failures.
7. Human Override Record: Summary statistics of human override rates for the preceding 12-month period, including trend analysis.

3.2 Additional MEP Components — AI/ML Pricing Models

For AI/ML systems used in credit pricing (Tier 1), the MEP must additionally include:

8. LLM and Agent Architecture: Description of the LLM backbone, agent orchestration framework, tool-calling design, and memory architecture. Version control information for all model components.
9. Pricing Floor Compliance Mechanism: Technical description of how the model enforces pricing floor constraints. Evidence that the system is incapable of recommending a rate below the applicable floor without generating a policy exception flag.
10. Data Grounding Evidence: Documentation of the RAG (Retrieval-Augmented Generation) architecture including: vector database configuration, document sources indexed, embedding model specifications, retrieval strategy, and re-indexing schedule.
11. Knowledge Graph Schema: For systems using a Knowledge Graph for rule lookup, documentation of the graph schema, node types, edge types, and data population procedures.
12. Explainability Artefact: A sample model output with the accompanying explanation, demonstrating that the explanation satisfies the requirements of CBUAE Circular 2024/BSE/047 Article 6.
13. Human-in-the-Loop Control Evidence: Technical description of the interrupt mechanism, approval workflow, and audit log structure. Confirmation that bypassing human approval is technically prevented.
14. Hallucination Risk Assessment: Assessment of the risk that the model may generate factually incorrect pricing information and the controls in place to detect and prevent such errors reaching end-users.

> **IMPORTANT:** The Data Grounding Evidence section (item 3 above) must demonstrate that all pricing floor values referenced by the model are sourced from the Knowledge Graph or RAG system, and not from the LLM's parametric knowledge. Parametric knowledge is not auditable and may not reflect current policy.

4. Ongoing Monitoring Requirements

4.1 Performance Monitoring

All production AI/ML pricing models must be monitored monthly against the following metrics:

Metric	Monitoring Frequency	Alert Threshold
Pricing floor breach rate	Daily	Any breach triggers immediate investigation
Human override rate	Monthly	> 20% override rate triggers model review
Recommendation acceptance rate	Monthly	< 60% acceptance rate triggers performance review

RAG retrieval freshness score	Weekly	Average freshness < 0.80 triggers re-indexing
Knowledge Graph staleness	Weekly	Any node with effective_date > 6 months old without review
LLM response latency	Daily	p95 > 5 seconds triggers infrastructure review
Explanation completeness score	Monthly	< 90% complete explanations triggers model review

5. Incident Reporting

A model incident is defined as any event in which an AI/ML model produces an output that: (a) breaches a pricing floor; (b) results in customer detriment; (c) triggers a regulatory notification; or (d) results in a financial loss attributable to model error.

All model incidents must be reported to the Model Risk Management team within 24 hours of detection, and to the CBUAE within 30 calendar days where the incident is material (threshold: financial impact > AED 1 million or regulatory breach).